

Where Every Number Came From

EV hosting capacity study, IEEE 123-bus distribution feeder. Each quantity used in the study is listed with its origin. Numbers are grouped by how they were obtained, from strongest provenance to weakest: specified by the published benchmark, measured from data, cited from a published source, derived by calculation, assumed, or produced by the solved model.

1. Specified by the IEEE 123 benchmark

Read directly out of the feeder files. Not chosen by this study, and not modified.

QUANTITY	VALUE	HOW IT WAS OBTAINED
Buses and nodes	132 / 278	Counted from the compiled circuit with <code>Circuit.NumBuses()</code> and <code>NumNodes()</code> . 71 buses are three-phase, 4 are two-phase and 57 are single-phase, which is why the node count is not simply three times the bus count.
Spot loads	91 objects	One <code>New Load</code> . line each in <code>IEEE123Loads.DSS</code> . Cross-checked against <code>dss.Loads.Count()</code> so that a load failing to construct would be caught rather than silently dropped.
Connected load	3,490 kW 1,770 kvar	Sum of the <code>kW=</code> and <code>kvar=</code> properties across the 91 load definitions. Only seven distinct kW values appear, and 78 of the 91 loads are either 20 or 40 kW, confirming these are specified round figures rather than metered data.
Load models	59 / 17 / 15	Counts of model 1 (constant power), model 2 (constant impedance) and model 5 (constant current). This mix is why the delivered load at nominal is 3,519.3 kW rather than the 3,490 kW nameplate sum: voltage-dependent loads draw more where voltage sits above 1.0 pu.
Lines	126 spans	Including switches, which the benchmark models as very short zero-impedance line segments.
Line codes	29	Impedance matrices in <code>IEEELineCodes.DSS</code> . None declares <code>normamps</code> , which is why conductor thermal limits were not assessed.
Regulators	4 (7 windings)	Transformer plus <code>RegControl</code> pairs. All are set <code>vreg=120</code> <code>band=2</code> with line drop compensation at <code>R=3</code> <code>X=7.5</code> . Those LDC settings are the mechanism behind the battery's effect on end-of-feeder voltage.
Capacitor banks	750 kvar	One 600 kvar three-phase bank at bus 83 and three 50 kvar single-phase banks at buses 88, 90 and 92. Modelled as fixed, not switched.
Voltage bases	4.16 / 0.48 kV	Declared by the master file. This study adds 72 kV for the substation transformer's high side.

2. Measured from data

Extracted from a public dataset by this study. Reproducible from the source file.

QUANTITY	VALUE	HOW IT WAS OBTAINED
Peak hour	17	AESO hourly load by area and region, November 2023 to December 2024. Filtered to the Edmonton region column, January 2024, weekdays only, with 1 January excluded as a statutory holiday. That leaves 22 days and 528 hours. Averaged by hour and normalised to the maximum.
Elevated window	16:00 to 20:00	The hours within 5% of the peak in the same averaged profile.
Edmonton region peak	1,964.8 MW	Hour 17 of the January 2024 weekday average. The single worst hour of that month was 2,215.6 MW during the 11 January cold snap, which was not used because the study models a typical winter day.
AESO shape depth	0.732	Minimum of the normalised Edmonton profile, at hour 3. Measured but deliberately not used. Regional load includes commercial and industrial customers and is far flatter than residential, so using it directly would have removed the evening peak the study depends on. The AESO data supplied the timing only.

3. Cited from published sources

Taken from documents outside this study. Each is attributable.

QUANTITY	VALUE	HOW IT WAS OBTAINED
Residential consumption	600 kWh/month	Alberta Market Surveillance Administrator, Residential Load Profiles, April 2004. Used as the starting point for the ADMD calculation in section 4.
Daily load structure	morning peak HE8, midday trough HE10 to HE16	Same MSA study. It presents profiles graphically rather than in tables, so it supplied the shape of the day but no numeric values.
Substation practice	30/40/50 MVA, 8 to 12 feeders	General distribution planning practice for urban substations. The three ratings are cooling stages: ONAN, ONAF stage 1 and ONAF stage 2. Used to justify the per-feeder share in section 5.
Alberta voltages	72 / 240 / 500 kV up to 25 kV	EPCOR, How power gets to your home. Transmission and distribution levels respectively. Used to set the 72 kV high side and to state that the 4.16 kV benchmark voltage is conservative.

4. Derived by calculation

Computed from numbers above. Each is reproducible by hand from its inputs.

QUANTITY	VALUE	HOW IT WAS OBTAINED
Average demand	0.83 kW	600 kWh per month, divided by 30 days, divided by 24 hours.
Load factor	0.605	Mean of the 24 load shape multipliers, divided by the maximum. Because the shape is normalised so the peak is 1.0, this is simply the mean. The 24 values sum to 14.52, and $14.52 / 24 = 0.605$. Note this is a property of the assumed shape, not an independent measurement.
ADMD	1.38 kW	0.83 kW divided by the load factor of 0.605. Load factor is the ratio of average to peak demand, so dividing by it converts an average consumption figure into a coincident peak demand figure. ADMD stands for After Diversity Maximum Demand.
Dwellings	2,513	3,490 kW divided by 1.38 kW gives 2,529. The code divides each of the 91 loads separately and rounds to whole dwellings, which gives 2,513. Individual loads work out to between 14 and 152 dwellings.
Vehicles at a given penetration	penetration $\times 2,513$	Chargers are placed by sampling without replacement from a pool of 2,513 entries, so the count is exact. 29% gives 734 vehicles, 51% gives 1,289, and 100% gives 2,513.
Installed charger kW	vehicles $\times 7.2$	At 30% penetration that is $754 \times 7.2 = 5,428.8$ kW of installed capacity, which exceeds the feeder's own 3,490 kW peak. Only a fifth of it draws at any one hour, which is what the charging coincidence profile represents.
Transformer loading	$\text{sqrt}(P^2 + Q^2) / \text{kVA}$	Apparent power through terminal 1 as a percentage of nameplate. Apparent power rather than real power, because windings heat according to current, which is indifferent to whether that current is doing useful work.
Overload profile	$(\text{loading} - 100) / 100 \times 5,000$	Loading percentage converted back to kVA above the limit, clipped at zero. Summed across the day it gives the energy above the limit. Its maximum gives the worst single hour.
Threshold crossings	linear interpolation	Between the two swept points that bracket the limit. The 10-point sweep grid sets the precision, not the arithmetic, which is why the thermal threshold was re-checked on a 1% grid between 25% and 30%.
Battery rating	900 kW / 2,400 kWh	Power comes from the worst single hour of overload at 50% penetration, which is 865 kVA. Energy comes from the 1,871 kWh above the limit that day, divided by the usable fraction of 90% and adjusted for discharge efficiency, giving about 2,300 kWh. Both rounded up.

5. Assumed

Chosen by this study. These are the numbers with no source behind them, and they are the ones an assessor should press on.

QUANTITY	VALUE	HOW IT WAS OBTAINED
Substation rating	5,000 kVA	The most consequential assumption in the study. Justified two ways: a 40 MVA bank divided across 8 feeders, and separately about 3.8 MVA implied by the 4.16 kV getaway conductor ampacity. Tested. 7.5 MVA moves the threshold from 29% to 91%.
Transformer impedance	8%	Typical range for a distribution substation transformer is 7 to 9%. Percent impedance is referenced to the unit's own base, which is why a larger transformer also improves voltage.
High side voltage	72 kV	Chosen to match Alberta transmission practice. Verified to have no numerical effect: 115 kV and 72 kV give identical results to seven decimal places, because the transformer impedance is on its own base.
Source setpoint	1.00 pu	Conservative. Real substations with load tap changers typically hold 1.02 to 1.03 pu. Minimum feeder voltage is highly sensitive to this: 0.98 pu would put the feeder in violation before any EVs are added.
Load shape trough	0.35	No source. Checked and found immaterial, because it only affects hours 1 to 5 and both limits bind at the evening peak.
Charger rating	7.2 kW	Standard residential Level 2.
Charging profile	24 values, peak 0.20	Synthetic, and the most consequential untested assumption. Follows uncontrolled charge on arrival behaviour, the same behavioural model the literature uses, but with an assumed arrival distribution rather than one derived from travel survey data. Calibrated so the values sum to 1.46 fleet-hours, giving 10.5 kWh per vehicle per day, consistent with roughly 50 km at 0.2 kWh/km. EV load at the binding hour scales directly with the 0.20 peak, so a value of 0.30 would move the threshold from about 29% to about 20%.
Load Vminpu	0.85	Lowered from the OpenDSS default of 0.95 on EV loads. Below Vminpu a constant-power load silently reverts to constant impedance. Measured on a test circuit: a 2,000 kW load at 0.71 pu drew only 1,083 kW at the default. That reversion covers the exact voltage range this study measures.
Battery siting	bus 150r	Feeder head, immediately downstream of the constrained transformer. A distributed fleet would also support voltage, but voltage is not the binding constraint.
Battery parameters	10% reserve, 93% each way	Round-trip efficiency of about 86%. Charging is set to hours 1 to 5 at 60% of rating, and the dispatch target to 98% of nameplate rather than 100%, leaving margin for the loss and power factor changes that shaving itself causes.
Random seeds	5 per level	Five charger placements at each penetration level. Spread proved small, under 0.5 percentage points of threshold, because at this model's resolution every one of the 91 nodes receives chargers even at low adoption.

6. Produced by the solved model

Outputs of the power flow, not inputs. Each is the value that satisfies Kirchhoff's laws, the element impedances and the load demands simultaneously, to within 0.0001 pu.

QUANTITY	VALUE	HOW IT WAS OBTAINED
Stock minimum voltage	0.9792 pu	At node 65.1, the end of a single-phase lateral. Before the substation transformer was added. Validated against the published IEEE benchmark solution.
Stock source power	3,615.2 kW 1,311.5 kvar	$S = 3,845.8$ kVA at a power factor of 0.940. Power balance closes: 3,519.3 kW delivered to loads plus 96.0 kW of losses equals 3,615.3 kW at the source, to within 0.1 kW.
Stock losses	96.0 kW	2.65% of delivered power, inside the 2 to 4% typical of primary distribution. This is a validation in itself, since it indicates the line impedances are physically sensible.
Baseline minimum voltage	0.9747 pu	At node 65.1, after the substation transformer was added, over load-serving buses only. The unfiltered minimum is 0.9615 pu at bus 150, the regulator input terminal, which has no customers connected to it.
Baseline loading	79.6%	$S = 3,977.5$ kVA through the 5,000 kVA transformer at hour 17. Higher than the stock source figure because the added transformer contributes its own losses and reactive absorption.
Peak migration	hour 17 to 18	Above 20% penetration the system peak moves from the household peak hour to the charging peak hour. Marginal loading per 10 points of penetration rises from 5.6 to about 8.2 as a result, which is the ratio of charging coincidence at the two hours, 0.20 divided by 0.14.
Thermal threshold	29.2% (29.1% refined)	Where peak loading first reaches 100%. Refined on a 1% grid: 99.9% at 29% and 100.7% at 30%.
Voltage threshold	64.3%	Where minimum voltage at a load-serving bus first reaches 0.95 pu. More than twice the thermal threshold, which is why this feeder is thermally limited.
Threshold with battery	51.3%	With 900 kW and 2,400 kWh peak shaving. At 50% penetration the peak falls from 117.3% to 98.9%. At 60% the battery is power-limited and only reaches 107.3%.
Sensitivity range	29% to 91%	Across the tested assumptions. 5 MVA gives 29.2%, 7.5 MVA gives 91.2%, and doubling the ADMD gives 41.9% while changing the vehicle count only from 734 to 731.

Reproducing any of these: the specified values are in `feeder/IEEE123/`, the derived values are computed in `src/`, and the model outputs are in `data/sweep_all.csv`. Running `python src/run_study.py` regenerates every figure in sections 4 and 6 in a few seconds.